

The Row Sums Were the Method, Not the Theorem: a Machine-Checked Chain from a Positive Weight to Exponential Decay of Correlations, and a Misattributed Uniformity Wall

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Abstract

A formal development of a two-dimensional spatial transfer kernel proves its uniform spectral statements by Schur’s test on *constant* row sums, and carries a theorem, labelled THE OBSTRUCTION in its own source, stating that those row sums stop being constant the moment the spatial coupling is switched on. That theorem obstructs the *method*. It was read as obstructing the *conclusion*, and the reading outlived its own retraction.

We prove, in Lean 4 with no **sorry** and no project axiom, a chain of four theorems. **First**, the one-bond influence envelope: flipping one neighbour across a bond of strength $J \geq 0$ moves a two-state site’s conditional by at most $\tanh J$, and by exactly $\tanh J$ at one field, so $\tanh J$ is the *least field-uniform* bound. It is an interdependence majorant, not the minimal coefficient of any particular finite model, whose reachable fields form a discrete set — a distinction Section 2 makes precisely. **Second**, those envelopes assembled into a matrix discharge every *matrix-side* hypothesis of the third (the two conditions on the ambient distance remain hypotheses), and turn a purely *local* covering condition into the remaining row-sum bound holding at every site of every volume — which is the coupling window, and which we compute: at a site carrying two bonds of strength β and two of strength γ it is exactly $2 \tanh |\beta| + 2 \tanh |\gamma|$, so the window is the hypothesis of a theorem rather than a sentence. **Third**, the resolvent bound: for a nonnegative matrix C over a finite index type, supported on pairs at distance at most one and with row sums *bounded* by $\alpha < 1$, without any constancy assumption,

$$\sum_{n < N} (C^n)_{ij} \leq \frac{\alpha^{\text{dist}(i,j)}}{1 - \alpha} \quad \text{for every } N,$$

with no reference of any kind to the cardinality of the index type. **Fourth**, *Dobrushin’s comparison estimate itself*, which an earlier version of this paper named as its one open analytic step: for a probability weight μ on a finite product space S^t , invariant under a family of single-site conditionals whose influence is dominated by C ,

$$|\text{Cov}_\mu(f, g)| \leq \frac{1}{4} \sum_{i,j} \delta_i(f) \left(\sum_n (C^n)_{ij} \right) \delta_j(g),$$

where δ_i is oscillation in the single coordinate i ; composing with the third theorem replaces the inner series by $\alpha^{\text{dist}(i,j)} / (1 - \alpha)$ and yields exponential decay of connected correlations at a rate read off single-site data. The constant $\frac{1}{4}$ is *attained*, on a witness inside the kernel. **Fifth**, and the reason the chain now begins where a reader’s data begins: from a strictly positive weight w on S^t *alone* we construct the measure $\mu_w = w/Z$ and its heat-bath kernel, *prove* that μ_w is invariant — by an involution on $S^t \times S$, so that step is an identity rather than

an estimate — and *prove* that the intrinsic Dobrushin matrix c^w has vanishing diagonal and dominates the influence. Every hypothesis of the fourth theorem is thereby discharged except one: $\sum_k c_{ik}^w \leq \alpha < 1$, Dobrushin’s condition itself, which is a condition on the interaction and not on the machinery. Each rung carries a non-vacuity witness and every declaration is oracled.

Separately, and by measurement rather than proof, we remove the wall from where it was being placed. For the coupled kernel the second-eigenvalue ratio $\text{specRatio}(L)$ saturates strictly below one in each of six cells tested inside Onsager’s disordered region — with the spatial weight switched on — and reaches one to six decimal places in the two ordered cells tested. The control at zero spatial coupling reproduces the *proved* identity $\text{specRatio} = \tanh \beta$ to six decimals, so the harness is calibrated against a theorem rather than against itself. These data are eight cells at $L \leq 12$ with an extrapolation. On the tested grid they *contradict* the blanket attribution of the degeneracy to the mere presence of the spatial weight; they are not, and are not offered as, a refutation of an asymptotic claim, since a sequence can appear to saturate at $L \leq 12$ and turn afterwards. They do *not* locate the transition, which would need a dense sweep and finite-size scaling that we do not attempt.

What is not proved. The volume-uniform bound on specRatio for the coupled kernel is *not* established here. The comparison estimate is proved for an *arbitrary* invariant weight with dominated influence; it is *not* instantiated at the Gibbs weight of the development, which needs its single-site conditionals constructed and their invariance and domination discharged, and that work is not done here. Dobrushin’s uniqueness theorem and his comparison estimate are classical [1, 2, 5]; nothing in this paper is new mathematics. What is offered is a mechanised, non-vacuous chain from a single bond to exponential decay of correlations, and measured counterevidence against an attribution. No consequence for Yang–Mills theory is stated, suggested, or implied.

1 What this paper claims, and what it does not

Claimed, machine-checked. Theorems 3.1, 3.2, 3.3, 4.3 and 5.2, their witnesses and the corollaries that compose them, in [DobrushinCoefficient.lean](#), [DobrushinRowSum.lean](#), [DobrushinMatrix.lean](#), [DobrushinOscillation.lean](#), [DobrushinGruss.lean](#), [DobrushinConditional.lean](#) and [DobrushinComparison.lean](#) and [DobrushinGibbs.lean](#). Every declaration is oracled and depends on exactly Lean’s three standard axioms.

Claimed, measured. The saturation table of Section 6. These are floating-point computations at finite extent with a documented extrapolation; they are evidence, not proof, and they are labelled as such wherever they appear.

Claimed, about the record. Section 2 states precisely what the source theorem `coupled_rowSums_not_constant` does and does not license. This is a statement about a formal development, and it is verifiable by reading that development.

Not claimed. That specRatio of the coupled kernel is bounded away from one uniformly in the extent: what is proved is decay of correlations of the Gibbs weight, and the transport into the operator formulation is untouched. That Dobrushin’s condition holds for any particular interaction — it is the single hypothesis Theorem 5.2 leaves to its user. That the window derived from Dobrushin’s condition is optimal — it is not, and Section 8 says by how much. That any of this bears on the Yang–Mills mass gap.

2 The setting, and the theorem that was read too widely

Fix an extent L and consider configurations $\sigma \in \{0,1\}^L$ of one slice. The development under discussion carries a time-bond kernel that factorises over sites,

$$K_\beta(\sigma, \tau) = \prod_{j < L} b_\beta(\sigma_j, \tau_j), \quad b_\beta(s, s') = e^{\pm\beta},$$

and an arbitrary strictly positive slice weight w , the Gibbs weight of a space-time configuration being the product of the slice weights and the time bonds. When w is itself a product over spatial bonds with coupling γ , the system is the anisotropic two-dimensional Ising model.

Two facts from that development matter here.

Proposition 2.1 (proved elsewhere in the development). *At $\gamma = 0$ and for every $L \geq 1$, $\text{specRatio} = \tanh \beta$ exactly.*

The proof is Schur’s test [8]: the decoupled kernel has *constant* row sums, and the loss on the odd part of an observable is exactly compensated by the odd bond eigenvalue at every step.

Proposition 2.2 (`coupled_rowSums_not_constant`). *At $\gamma > 0$ the row sums of the coupled kernel are not constant: two explicit configurations of a two-site slice have different row sums.*

Proposition 2.2 is labelled THE OBSTRUCTION in the source. *It obstructs Schur’s test.* It says nothing whatever about `specRatio`, it is proved on a two-site witness, and no statement quantified over L appears in it.

A stronger reading — that the spatial weight destroys uniformity — was entered into the development and then *retracted of record* as unproved, leaving behind only a measurement of degeneracy.

What is documented, and can be checked. The retraction of the sentence from the manuscript is recorded at Addendum 561 of the development’s verification ledger; its removal from the module docstring, which the first retraction had missed, at Addendum 563; and its survival in a hand-typed submission form, caught only because the form was read in full before delivery, at Addendum 568. Three removals were required for one sentence, and the third was the one that would have been published.

What is inference, and is labelled as such. We observe that no statement quantified over L for the coupled kernel was attempted in the interval, and we attribute this to the retracted reading continuing to govern target selection. That attribution is ours; it is not established by a diff, and no artefact in the development records a decision not to attempt something. The verifiable half of the observation is the narrower one, and it is enough for this paper: the guards that caught the sentence three times compare identifiers and substrings inside files, so whatever else was carrying the reading was not a file.

3 The chain

Three theorems, each machine-checked, composing into one statement. The first is about a single bond, the second assembles bonds into a matrix, the third is about that matrix and never sees the model.

3.1 The one-bond influence envelope, and that it is least among field-uniform bounds

A two-state site in a real field h has conditional $P(+1) = (1 + \tanh h)/2$, written from the Boltzmann weights and nothing else. Flipping one neighbour across a bond of strength J moves that field to $h - 2J$. Write $\text{tv}(h, h') = |P_h(+1) - P_{h'}(+1)|$.

Theorem 3.1 (machine-checked). *For every $J \geq 0$,*

$$\text{tv}(h, h - 2J) \leq \tanh J \quad \text{for all } h, \quad \text{and} \quad \text{tv}(J, -J) = \tanh J.$$

Hence $\tanh J$ is the least upper bound of $\{\text{tv}(h, h - 2J) : h \in \mathbb{R}\}$.

Proof. $\tanh a - \tanh b = \sinh(a - b)/(\cosh a \cosh b)$, and $\cosh a \cosh b = (\cosh(a + b) + \cosh(a - b))/2 \geq (1 + \cosh(a - b))/2$ since $\cosh \geq 1$. With $a = h$ and $b = h - 2J$ the difference of the fields is $2J$, so the denominator is at least $(1 + \cosh 2J)/2 = \cosh^2 J$ while the numerator is $\sinh 2J = 2 \sinh J \cosh J$. Dividing by two, $\text{tv} \leq \sinh J \cosh J / \cosh^2 J = \tanh J$. At $h = J$ the two fields are $+J$ and $-J$, their sum is zero, $\cosh 0 = 1$, and every inequality above is an equality. $\square$

What this is, stated carefully, because the obvious reading of it is wrong. The supremum is taken over *all* $h \in \mathbb{R}$. In a concrete finite system with the remaining couplings and any external field fixed, the effective fields actually reachable at a site form a discrete subset of $\mathbb{R}$, and the value $h = J$ at which equality holds need not correspond to any admissible boundary condition of that system. So the true Dobrushin coefficient

$$c_{ij} = \sup\{\|\mu_i(\cdot | \eta) - \mu_i(\cdot | \eta')\|_{\text{TV}} : \eta, \eta' \text{ differ only at } j\}$$

of a particular model may satisfy $c_{ij} < \tanh |J_{ij}|$ strictly.

Theorem 3.1 is therefore a **sharp field-uniform one-bond influence envelope**: sharp as a bound over arbitrary real fields, and a valid interdependence majorant, $c_{ij} \leq \tanh |J_{ij}|$, for every model with that bond. That is all Dobrushin's condition needs and it invalidates nothing downstream — but it is not the claim that $\tanh |J|$ is the minimal interdependence matrix of any particular system, and this paper does not make that claim. The window below is a sufficient condition, explicit and not optimal.

The attainment still earns its place: without it the envelope would be some number above the influence, and one could not say the envelope is the best field-uniform one. In Lean the three statements are `tvField_le`, `tvField_attained` and `tvField_isLUB`.

3.2 The matrix, and the window as a hypothesis

Given sites ι , bond strengths J and a distance d , put

$$C_{ij} = \begin{cases} \tanh |J_{ij}| & i \neq j \text{ and } d(i, j) \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 3.2 (machine-checked). *C is entrywise nonnegative, vanishes on the diagonal, and vanishes whenever $d(i, j) > 1$. Consequently every matrix-side hypothesis of Theorem 3.3 — (i) and (ii) — holds by construction. Furthermore, if at every site the influence support is contained in a finite set whose bond coefficients sum to at most α , then the row-sum hypothesis*

holds at every site, with neither the statement nor α mentioning the cardinality of the index type. Finally, at a site carrying two bonds of strength β and two of strength γ ,

$$\sum_j C_{ij} = 2 \tanh |\beta| + 2 \tanh |\gamma|.$$

Three remarks, in decreasing order of how easily the statement is over-read.

What is not discharged. The construction settles the matrix-side hypotheses. It does *not* discharge the two conditions on the ambient distance — $d(i, i) = 0$ and the triangle inequality — which Corollary 3.7 still takes as hypotheses, as does the Lean endpoint. For a graph distance they are immediate; for an arbitrary d they are assumptions, and the paper does not pretend otherwise.

One site versus a family. The last display is arithmetic at one site of one five-point graph, and by itself it would say nothing about a family of volumes. The middle clause is what makes it say something: it converts a *local* geometric fact — each site’s influence support is covered by a set whose coefficients sum below one — into the row-sum hypothesis holding simultaneously at every site of every volume with that local geometry. In Lean these are `rowSum_bound_of_local_cover` and `dobrushin_bound_of_local_cover`.

The window. It has been carried in this lane’s prose as $2 \tanh \beta + 2 \tanh \gamma < 1$; it is now computed rather than asserted and is consumed by Corollary 3.7. It remains a sufficient condition and, by Section 8, a conservative one. The distance used in the computation is the graph distance on a star, proved reflexive at zero and triangular by exhaustive checking over the 125 triples.

3.3 The resolvent bound

Theorem 3.3 (machine-checked). *Let ι be a finite type, $d : \iota \times \iota \rightarrow \mathbb{N}$ satisfy $d(i, i) = 0$ and the triangle inequality, and let $C : \iota \times \iota \rightarrow \mathbb{R}$ satisfy*

- (i) $C_{ij} \geq 0$ for all i, j ;
- (ii) $C_{ij} = 0$ whenever $d(i, j) > 1$;
- (iii) $\sum_j C_{ij} \leq \alpha$ for all i , with $0 \leq \alpha < 1$.

Then for all i, j and every $N \in \mathbb{N}$,

$$\sum_{n \in [0, N)} (C^n)_{ij} \leq \frac{\alpha^{d(i, j)}}{1 - \alpha},$$

and consequently $\sum_n (C^n)_{ij} \leq \alpha^{d(i, j)} / (1 - \alpha)$.

Proof. Three steps, each an induction on n . Nonnegativity is preserved by powers. Row sums multiply: expanding (C^{n+1}) and exchanging the two finite sums, $\sum_j (C^{n+1})_{ij} = \sum_k (C^n)_{ik} \sum_j C_{kj} \leq \alpha \sum_k (C^n)_{ik} \leq \alpha^{n+1}$, so each entry, being at most its row sum, is at most α^n . Support: if $n < d(i, j)$ then $(C^n)_{ij} = 0$, since in $(C^{n+1})_{ij} = \sum_k (C^n)_{ik} C_{kj}$ each k has either $d(i, k) > n$, killing the first factor by induction, or $d(i, k) \leq n$, whence $d(i, j) \leq d(i, k) + d(k, j) \leq n + d(k, j)$ forces $d(k, j) > 1$ and kills the second. Combining, $(C^n)_{ij} \leq \alpha^n$ when $n \geq d(i, j)$ and vanishes otherwise; splitting $[0, N)$ at $d(i, j)$ and reindexing the surviving block by $n = d(i, j) + m$ gives $\alpha^{d(i, j)} \sum_{m < N - d(i, j)} \alpha^m$, and the geometric sum is at most $(1 - \alpha)^{-1}$. $\square$

Remark 3.4 (where the volume-freeness is). *The cardinality of ι appears nowhere in the hypotheses or the conclusion. That is the entire content of the theorem: the index type may be as large as one likes and neither α nor the exponent notices. Every constant is read off single-site data.*

Remark 3.5 (the hypothesis that matters here). *Hypothesis (iii) asks for row sums bounded by α . It never asks for them to be constant. Proposition 2.2 refutes constancy and leaves (iii) untouched: it is precisely the crack through which the coupled kernel passes after Schur's test has died.*

Proposition 3.6 (non-vacuity, machine-checked). *On the three-point chain with d the graph distance and $C_{ij} = 1/4$ for $d(i, j) \leq 1$ and 0 otherwise, hypotheses (i)–(iii) hold with $\alpha = 3/4 < 1$, the matrix is not the zero matrix, the two endpoints are at distance exactly two, and the conclusion reads: the whole resolvent series between the endpoints is at most $9/4$.*

The witness is not decoration. Hypotheses (ii) and (iii) are jointly satisfiable by the zero matrix, for which the conclusion is trivial, and by data with $d \equiv 0$, for which the exponent is trivial. Proposition 3.6 excludes both.

3.4 The composition

Corollary 3.7 (machine-checked). *Let J be bond strengths on a finite set of sites with a distance d satisfying $d(i, i) = 0$ and the triangle inequality, and let C be the matrix of Theorem 3.2. If every row sum of C is at most $\alpha < 1$, then for all i, j and every N ,*

$$\sum_{n < N} (C^n)_{ij} \leq \frac{\alpha^{d(i,j)}}{1 - \alpha},$$

and the same holds for the full series.

This is the whole chain in one line: the envelope of Theorem 3.1 enters the matrix of Theorem 3.2, which satisfies the hypotheses of Theorem 3.3, whose conclusion mentions no cardinality. In Lean it is `dobrushin_resolvent_bound`, and its proof is the application of one theorem to three others.

4 The comparison estimate

Theorems 3.1–3.3 bound a resolvent. Nothing in them mentions a measure, and so nothing in them says that correlations are controlled by that resolvent. That implication is Dobrushin's comparison estimate, and the previous version of this paper named it as the single open analytic step. This section proves it.

4.1 The objects

Let ι be a finite index type and S a finite nonempty state space, so that the configuration space S^ι is finite. For $f : S^\iota \rightarrow \mathbb{R}$ put

$$\delta_i(f) = \max\{|f(\eta) - f(\eta')| : \eta, \eta' \text{ agree off } i\},$$

the oscillation of f in the single coordinate i . A *single-site kernel* is a family $p_i(\eta, \cdot)$ of probability weights on S that is *local*: $p_i(\eta, \cdot)$ does not depend on η_i . It acts on observables by

$$(E_i f)(\eta) = \sum_{s \in S} p_i(\eta, s) f(\eta^{i \rightarrow s}),$$

where $\eta^{i \rightarrow s}$ is η with the value at i replaced by s . We say a matrix C *dominates the influence* of the kernel if, for $k \neq i$,

$$\|p_i(\eta, \cdot) - p_i(\eta', \cdot)\|_{\text{TV}} \leq C_{ik} \quad \text{whenever } \eta, \eta' \text{ agree off } k.$$

The orientation is the one fixed before any of this Lean was written: the *first* index of C is the site being conditioned, the *second* is the site whose value moved. On a symmetric cell the two conventions are indistinguishable, which is why fixing it in advance mattered; the gate that discriminates is described at the end of this section.

4.2 The key lemma

Lemma 4.1 (machine-checked). *Let p be a local single-site kernel and C dominate its influence. Then for every observable f :*

$$\delta_i(E_i f) = 0, \quad \delta_k(E_i f) \leq \delta_k(f) + C_{ik} \delta_i(f) \quad (k \neq i).$$

Proof. The first clause is semantics, not estimation: by locality $E_i f$ does not depend on the coordinate i at all. For the second, split the difference of $E_i f$ at two configurations agreeing off k into a *transport* term, where the kernel is held fixed and f is evaluated at two configurations that still agree off k — here $k \neq i$ is load-bearing — and a *kernel variation* term $\sum_s (p_i(\eta, s) - p_i(\eta', s)) f(\eta^{i \rightarrow s})$. The first is at most $\delta_k(f)$ because $p_i(\eta, \cdot)$ is a probability weight. The second is a signed mass of total zero tested against a section of f along coordinate i ; a zero-sum mass cannot see a constant, so subtracting the midpoint of the range gives $|\sum_s a_s g_s| \leq \frac{1}{2} (\sum_s |a_s|) \text{osc}(g)$, which is total variation times oscillation, at most $C_{ik} \delta_i(f)$. $\square$

The three statements are kept apart in the source on purpose. The vanishing clause is `deltaAt_condExp_self` and mentions no matrix; the transport clause is `deltaAt_condExp_le` and asks only that C *dominate*, never that it be minimal — the same distinction that cost this lane two versions over $\tanh|J|$; and their assembly into a single vector inequality, `deltaAt_condExp_le_matrix`, is the *only* statement that assumes $C_{ii} = 0$. That the zero diagonal is a declared hypothesis of the matrix form and not an accident of the physical cell was decided, and written into the lane charter, before this module existed.

4.3 The one-step quarter

Lemma 4.2 (machine-checked). *For a probability weight p on a finite set and any f, g with ranges in $[m_1, M_1]$ and $[m_2, M_2]$,*

$$|\mathbb{E}_p[fg] - \mathbb{E}_p[f] \mathbb{E}_p[g]| \leq \frac{1}{4} (M_1 - m_1)(M_2 - m_2),$$

and the constant $\frac{1}{4}$ is attained.

Proof. Not Cauchy–Schwarz. The centred mass $a_x = p_x(f_x - \mathbb{E}_p f)$ has total zero, so the signed-mass inequality above gives $|\text{Cov}| \leq \frac{1}{2} (\sum_x |a_x|) \text{osc}(g)$, and $\sum_x |a_x|$ is the mean absolute deviation of f about its mean, which is at most $(M_1 - m_1)/2$: split it into its positive and negative parts, observe they are *equal* because the mass is centred, bound one by $(M_1 - \mathbb{E}f)T$ and the other by $(\mathbb{E}f - m_1)(1 - T)$ with T the mass above the mean, and multiply. The fair two-point weight with $f = g$ an indicator attains it. $\square$

Each factor $\frac{1}{2}$ is born in exactly one lemma and the quarter is their product. Popoviciu’s variance bound, also machine-checked in the same module, is *not* used: it is the L^2 companion, kept separate so that a difficulty in either could not block the other.

4.4 The estimate

Theorem 4.3 (machine-checked). *Let μ be a probability weight on S^v , invariant under every E_i (that is, $\mathbb{E}_\mu[E_i F] = \mathbb{E}_\mu[F]$ for all F and i), let C dominate the influence of the kernel with $C_{ii} = 0$ and row sums at most $\alpha < 1$. Then for all observables f, g ,*

$$|\text{Cov}_\mu(f, g)| \leq \frac{1}{4} \sum_{i,j} \delta_i(f) \left(\sum_n (C^n)_{ij} \right) \delta_j(g),$$

and, composing with Theorem 3.3 when C is supported at distance at most one,

$$|\text{Cov}_\mu(f, g)| \leq \frac{1}{4(1-\alpha)} \sum_{i,j} \delta_i(f) \alpha^{\text{dist}(i,j)} \delta_j(g).$$

For observables depending on single sites i_0 and j_0 this reads

$$|\text{Cov}_\mu(f, g)| \leq \frac{\delta_{i_0}(f) \delta_{j_0}(g)}{4(1-\alpha)} \alpha^{\text{dist}(i_0, j_0)} :$$

exponential decay in the distance, at a rate free of the volume.

Proof. Let $P = \frac{1}{m} \sum_i E_i$ be the random-scan average, $m = \#\iota$. It preserves μ -expectations, so telescoping over N steps,

$$\text{Cov}_\mu(f, g) = \sum_{n < N} \left(\text{Cov}_\mu(P^n f, g) - \text{Cov}_\mu(P^{n+1} f, g) \right) + \text{Cov}_\mu(P^N f, g).$$

Each increment is $\mathbb{E}_\mu[g \cdot (h - Ph)]$ with $h = P^n f$; averaging over the site chosen and using invariance once more, the i -th contribution equals $\mathbb{E}_\mu[E_i(g(h - E_i h))]$, whose integrand is *pointwise* the covariance of the two sections of g and h along coordinate i under $p_i(\eta, \cdot)$. By Lemma 4.2 and the fact that a section oscillates at most δ_i , each increment is at most $\frac{1}{4m} \sum_i \delta_i(g) \delta_i(P^n f)$.

By Lemma 4.1 the oscillation vector of the iterates is transported by $v \mapsto (1 - \frac{1}{m})v + \frac{1}{m}C^\top v$, so $\delta(P^n f)$ is dominated coordinatewise by the n -th iterate of that map applied to $\delta(f)$. Two things follow, both by finite induction and neither taking a limit. Its ℓ^1 norm contracts by $1 - \frac{1-\alpha}{m}$ per step, so the telescoping remainder $|\text{Cov}_\mu(P^N f, g)|$, bounded crudely by $\frac{1}{4} \text{osc}(g) \sum_i \delta_i(P^N f)$, dies geometrically. And the *accumulated* transport $\sum_{n < N}$ of those iterates is dominated entrywise by m times the N -th partial sum of $\sum_n (C^\top)^n$ — which is the step where the factor m introduced by the averaging is exactly cancelled, leaving a bound with no m in it. Assembling gives the finite- N statement; letting $N \rightarrow \infty$ in the *bound* gives the series form. $\square$

Every statement up to the last sentence of that proof is a finite inequality over a finite configuration space. The single limit is taken at the level of the bound, which is a design decision this time recorded before the module existed, so that no infinite-volume object is constructed anywhere.

Proposition 4.4 (non-vacuity and attainment, machine-checked). *One site, two states, the fair kernel, the uniform weight, $C = 0$, $\alpha = 0$, $d \equiv 0$. Every hypothesis of Theorem 4.3 holds by computation, and on the indicator observable the two-point conclusion is an equality: $|\text{Cov}| = \frac{1}{4}$ against a bound whose value is $\frac{1}{4}$.*

The witness does two jobs. It shows the hypotheses are jointly satisfiable — invariance of μ under all E_i is a real constraint, not a formality. And because the bound is *attained* there, the constant $\frac{1}{4}$ cannot be lowered by any argument, in this or any other proof of the same inequality.

4.5 The gates, and the one that discriminated

Five gates were registered before the corresponding Lean and are in [judge_dobrushin_d3.py](#) and [judge_dobrushin_d3b.py](#). Two are worth naming.

The gate that could have killed the rung. Lemma 4.1 was checked by brute force on small Ising cells over *all* $2^{|S^c|}$ Boolean observables — 256 of them at three sites — computing both sides exactly for every site pair. The gate reports the worst value of lhs – rhs, and a positive value would have refuted the lemma before a line of it was written. Measured: $+1.11 \times 10^{-16}$, one unit in the last place of a float64 comparison at scale one.

The gate that discriminated the orientation. On an Ising cell the minimal influence coefficients are symmetric, so $C = C^\top$ and a transposed index would pass every other gate and surface much later. A three-site cell built from eight distinct positive weights has $\max_{ij} |C_{ij} - C_{ji}| > 0$, and there the declared orientation passes with zero violations while the transposed one **fails with** 24. The convention in Lemma 4.1 is therefore the one the lemma needs, and that is established by measurement rather than by assertion.

Both certifiers were run in `normal` and in `optimized` (`python -O`) mode on the Linux plane, all four runs exiting zero with their check counters matching.

5 The instantiation: from a weight, not from data

Theorem 4.3 takes *data* — a weight, a kernel, a majorant — and six hypotheses. A reader is entitled to ask whether that data exists for anything, and whether the hypotheses are met by the object one actually has, which is an interaction, not a kernel. This section removes the question. Its input is a single function.

Definition 5.1. Let $w : S^t \rightarrow \mathbb{R}$ be strictly positive. Its *Gibbs weight* is $\mu_w = w/Z$ with $Z = \sum_{\eta} w(\eta)$, and its *heat-bath kernel* is

$$p_i^w(\eta, s) = \frac{w(\eta^{i \rightarrow s})}{\sum_{t \in S} w(\eta^{i \rightarrow t})}.$$

Its *intrinsic Dobrushin matrix* is

$$c_{ik}^w = \max \left\{ \|p_i^w(\eta, \cdot) - p_i^w(\eta', \cdot)\|_{\text{TV}} : \eta, \eta' \text{ agree off } k \right\},$$

the maximum over the finite configuration space — the coefficient itself, not a majorant chosen from outside.

Theorem 5.2 (machine-checked). *For every strictly positive w : μ_w is a probability weight; p^w is nonnegative, normalised and local; μ_w is invariant under every E_i^w ; $c_{ii}^w = 0$; and c^w dominates the influence of p^w . Consequently, if $\sum_k c_{ik}^w \leq \alpha < 1$ at every site, then for all observables*

$$|\text{Cov}_{\mu_w}(f, g)| \leq \frac{1}{4} \sum_{i,j} \delta_i(f) \left(\sum_n (c^w)_{ij}^n \right) \delta_j(g),$$

and if moreover c^w vanishes beyond distance one for a distance d , then for single-site observables at i_0 and j_0 ,

$$|\text{Cov}_{\mu_w}(f, g)| \leq \frac{\delta_{i_0}(f) \delta_{j_0}(g)}{4(1 - \alpha)} \alpha^{d(i_0, j_0)}.$$

Proof. Normalisation and locality are computations: the local partition sum $\sum_t w(\eta^{i \rightarrow t})$ is positive, and it does not see η_i because $\eta^{i \rightarrow t}$ does not.

Invariance is the only step with content, and it is an identity rather than an estimate. Write $\mathbb{E}_{\mu_w}[E_i^w F]$ as a sum over $S^t \times S$ and apply the involution

$$\tau_i(\eta, s) = (\eta^{i \rightarrow s}, \eta_i), \quad \tau_i \circ \tau_i = \text{id}.$$

Under τ_i the integrand $\mu_w(\eta) p_i^w(\eta, s) F(\eta^{i \rightarrow s})$ becomes $\mu_w(\eta^{i \rightarrow s}) \cdot \frac{w(\eta)}{\sum_t w(\eta^{i \rightarrow t})} \cdot F(\eta)$, using that the local partition sum is unchanged along the fibre. Summing that over s first reconstitutes $\sum_s \mu_w(\eta^{i \rightarrow s})$, which is the local partition sum over Z ; it cancels the denominator exactly, leaving $\mu_w(\eta) F(\eta)$. Reindexing a finite sum by an involution is free, and that is the whole proof.

The zero diagonal is a theorem, not a hypothesis: c_{ii}^w is a maximum of total variations between $p_i^w(\eta, \cdot)$ and $p_i^w(\eta^{i \rightarrow s}, \cdot)$, which are equal by locality. Domination is immediate because c^w is defined as the maximum. The conclusions are Theorem 4.3 and its corollaries applied to this data. $\square$

Three remarks, in the order in which the statement is most easily over-read.

What is left as a hypothesis, and it is exactly one thing. Dobrushin's condition $\sum_k c_{ik}^w \leq \alpha < 1$ — plus, for the distance form, that c^w has range one. Both are conditions on the *interaction*, not on the machinery: nothing about kernels, invariance or majorants survives for a user to discharge. In Lean the endpoints are `gibbs_covar_le_resolvent`, `gibbs_covar_exp_decay` and `gibbs_covar_two_point`.

Why the heat-bath kernel and not another. The comparison estimate holds for *any* invariant local kernel; the heat-bath one is chosen because its invariance is an involution rather than a computation, and because its influence coefficients are the ones Dobrushin's condition is usually stated for. Nothing downstream depends on the choice.

The intrinsic matrix versus the envelope. c^w is the coefficient, so for a bonded interaction it is bounded by the envelope $\tanh |J_{ij}|$ of Theorem 3.1 and may be strictly smaller. Gate G14 measures that gap on registered cells; the paper does not prove the comparison, because Theorem 5.2 does not need it — a user with an interaction may apply the envelope to *bound* $\sum_k c_{ik}^w$, and that is the only role Theorem 3.1 plays here.

Proposition 5.3 (non-vacuity of the instantiation, machine-checked). *For the uniform weight on one site with two states, every hypothesis of Theorem 5.2 holds by computation, its Gibbs weight is the witness weight of Proposition 4.4, and the instantiated two-point bound is again an equality at $\frac{1}{4}$.*

So the instantiated hypotheses are jointly satisfiable and the instantiated bound is still the sharp one. This matters more than the usual non-vacuity check, because invariance is a genuine constraint on the pair (weight, kernel) and a construction that failed it would make Theorem 5.2 say nothing.

Gates. `judge_dobrushin_d4.py`, registered before this Lean. G12 checks invariance as an *identity* to 10^{-12} on deliberately asymmetric weights, exhaustively over all Boolean observables and all sites — a residual above roundoff would have killed the involution argument before it was written. G13 checks that the heat-bath conditional of an Ising weight is $(1 + \tanh h)/2$ at the local field. G14 checks that the intrinsic coefficients are dominated by $\tanh |J|$ entrywise, which is the measured content of “the envelope is a majorant, not the minimum” at the level of the measure.

6 The measurement, and where the wall is

This section is numerical. Nothing in it is proved, and nothing in it is used by Section 3.

Let $T_L = W_\gamma^{1/2} K_\beta W_\gamma^{1/2}$ be the symmetrised coupled kernel on $\{\pm 1\}^L$ with free spatial boundary, and $r(L) = |\lambda_1|/|\lambda_0|$ its second-eigenvalue ratio, which is `specRatio` of the development. We compute $r(L)$ for $L = 2, \dots, 12$ by dense symmetric eigendecomposition and extrapolate by Aitken.

β	γ	$\sinh 2\beta \sinh 2\gamma$	$r(2)$	$r(12)$	$r(\infty)$
<i>controls at zero spatial coupling</i>					
0.20	0.00	0.0000	0.197375	0.197375	$\tanh \beta = 0.197375$
0.40	0.00	0.0000	0.379949	0.379949	$\tanh \beta = 0.379949$
<i>disordered</i>					
0.10	0.10	0.0405	0.1096	0.1209	0.1214
0.20	0.20	0.1687	0.2360	0.2895	0.2922
0.30	0.30	0.4053	0.3733	0.5127	0.5230
0.40	0.40	0.7887	0.5122	0.7813	0.8162
0.10	0.80	0.4783	0.1651	0.4013	0.4570
0.80	0.10	0.4783	0.7246	0.8031	0.8075
<i>ordered</i>					
0.60	0.60	2.2785	0.7489	0.9992	1.000003
0.80	0.80	5.6433	0.8893	0.999999	1.000000

$r(\infty)$ is an Aitken extrapolation of the last three extents. Values above one in the ordered rows are extrapolation overshoot at the sixth decimal and are reported as measured.

Each of the six cells tested inside Onsager’s disordered region, $\sinh 2\beta \sinh 2\gamma < 1$, saturates strictly below one, with increment ratios near 0.6; both ordered cells tested reach one to six decimal places. The control at $\gamma = 0$ reproduces $\tanh \beta$ to six decimals, which is Proposition 2.1 — the harness is calibrated against a theorem of the development rather than against itself.

What these data support, stated at their strength and no further. They are consistent with the degeneracy being a property of the *ordered* phase, and they are *not* consistent with the degeneracy being caused by switching on the spatial weight: six disordered cells with the weight switched on fail to degenerate. That second, negative reading is the one this section is for, and it is the one that reopened the question.

What they do not support. They do not locate the transition. Eight cells at $L \leq 12$, with a three-term Aitken extrapolation whose overshoot is visible in the ordered rows above, cannot place a phase boundary; doing so would require a dense sweep near $\sinh 2\beta \sinh 2\gamma = 1$, finite-size scaling, and control of the extrapolator, none of which is attempted here. A reader should take from this section that the wall is *not* where the development had been treating it as being, and nothing about where it is.

Gates. Three gates were registered in [scripts/judge_dobrushin.py](#) together with the lane charter, before a line of the Lean of Section 3 was written. Each exits non-zero on failure; none bundles a theorem with its witness; the bound gate predicts a number rather than sampling. One of them was later found not to exercise the hypothesis of Remark 3.5 — it normalised every row to *exactly* α — and was redesigned to carry unequal row sums, strictly harder, declared in its own docstring and committed before being run. The gates emit a verdict only after an explicit counter confirms that the expected number of checks actually executed, because an empty failure list is also what zero checks produce. No acceptance decision anywhere in this lane depends on a Python `assert`.

7 What remains, named exactly

Theorems 3.1–5.2 are an engine, not a conclusion. Turning them into a volume-uniform statement about `specRatio` requires, in order:

1. a valid single-site interdependence majorant, and the sufficient window it produces. **Done:** Theorems 3.1 and 3.2, machine-checked, with the window computed rather than asserted, and holding at every site of every volume with the given local geometry. Not done, and not needed: the *minimal* interdependence matrix of any particular system;
2. Dobrushin’s comparison estimate [2, 5] — that connected correlations are controlled by the resolvent of C , hence decay exponentially at a rate free of the extent. **Done:** Theorem 4.3;
3. construction of the single-site conditionals of a Gibbs weight, and discharge of every hypothesis that estimate consumes — invariance above all, which is a genuine constraint on the pair and not a formality. **Done:** Theorem 5.2, from a strictly positive weight alone, leaving Dobrushin’s condition on the intrinsic matrix as the single hypothesis a user supplies;
4. transport of that decay into the operator formulation: from correlations of the Gibbs weight to `specRatio` of the transfer kernel, through the finite-time interface described below. **Open, and it is the only open step in this list.**

What changed, stated at its exact size. An earlier version of this paper named step 2 as the only currently identified new analytic estimate and the bottleneck of the chain; a later one proved it, for arbitrary invariant data, and left step 3 open. Both are now closed. The honest description of what remains is that step 4 is not an estimate but a *translation*: from a statement about a measure to a statement about an operator. The vacuum pairing is an infinite-time object, so that translation must run through finite time extent with explicit boundary control, and nothing here shortens it. What this paper does claim is that between a positive weight and exponential decay of its correlations there is no longer a gap: the chain is closed end to end, and the only input a user supplies is Dobrushin’s condition.

The consumer, named. It is `volumeUniformGap` in [YangMills/OS/TransferGap.lean](#), and its signature, read off the elaborated declaration rather than from memory, is: for a family of real inner-product spaces H_i indexed by ι , transfer operators T_i and vectors Ω_i with `VacuumTransfer` $T_i \Omega_i$ for every i , and a rate $0 < r < 1$, the hypothesis

$$\forall i, \forall v \in H_i, \exists C, \forall n, \quad |\text{connCorr } T_i \Omega_i v n| \leq C r^n$$

yields $\exists m > 0$ with $\|\text{projectedTransfer } T_i \Omega_i\| \leq e^{-m}$ for every i . It takes, over a family of finite volumes, decay at a *common* rate and returns a *common* mass. Its quantifier order is worth stating because it is load-bearing: the prefactor C may depend on the volume and on the observable, and only r is shared. The development’s own standing objection — that a bound $C(L)\rho^N$ with $C(L) \rightarrow \infty$ says nothing — therefore does not apply to it, the conclusion being an operator-norm bound.

A design decision is recorded here rather than discovered later: the vacuum pairing is an infinite-time object, so step 3 must proceed through finite time extent with explicit boundary control, taking the limit only at the level of the bound. No infinite-volume state is constructed anywhere in this paper, and the word *clustering* is accordingly not used for one.

8 Prior art, and the exact delta

Dobrushin’s uniqueness theorem is classical [1, 2, 5] and nothing in this paper is new mathematics. We locate the contribution precisely.

Not explicit constants. The cell here is the anisotropic two-dimensional Ising model, whose transfer spectrum has been available in closed form since Onsager [6] and Kaufman [7]. A constant obtained from Dobrushin’s condition is *worse* than the exact answer — on the cells we measured, by a factor between roughly two and thirty-six. Any claim to have improved a constant for this model would be false, and we make none.

Generality, instead. Dobrushin’s coefficient is read off the single-site conditionals of an *arbitrary* positive finite-range slice weight, which is what the development’s Gibbs weight quantifies over, and where the exact solution is silent.

Mechanisation. A search conducted on 1 August 2026 over the public proof-assistant repositories and the arXiv listings for Lean, Coq and Isabelle returned no mechanisation of the finite-lattice Dobrushin interdependence matrix or of the comparison estimate. We state it that narrowly on purpose: the name *Dobrushin* attaches to several distinct results, and mechanised work exists that uses others of them, so a blanket claim of first formalisation would be both unsupported by the search we ran and ambiguous about which theorem it referred to. The same search does return a Lean 4 formalisation of the transverse-field Ising chain [11], with momentum-mode decomposition and a Jordan–Wigner transformation; it concerns the quantum chain rather than the classical two-dimensional transfer matrix, and displaces nothing here, but it means a blanket claim that the Ising model has not been formalised would not survive review, and we make none. A formalisation of the Perron–Frobenius theorem for irreducible nonnegative matrices also exists [10], as does a mechanisation of the Osterwalder–Schrader axioms for the Gaussian free field [12]. Reflection positivity and finite-lattice spectra for related models have been treated without mechanisation [13].

9 Limitations

All five theorems are elementary. Theorem 3.3 is three inductions; Theorem 3.1 is a hyperbolic identity and a bound on a denominator; Theorem 3.2 is a construction and a finite computation; Theorem 4.3 is a telescoping sum, one pointwise application of a ninety-year-old covariance bound, and two finite inductions; Theorem 5.2 is a reindexing by an involution. Their value is that they are machine-checked, non-vacuous, composed, and carry the hypothesis the dead method violated — not that any of them is deep.

Section 6 is numerics at $L \leq 12$ with an Aitken extrapolation. It provides measured counterevidence against an attribution and contradicts it on the tested grid; it does not prove where the wall is, and an extrapolation can mislead near a transition. The window is not optimal and is never called sharp.

Step 4 of Section 7 is open, and until it closes the volume-uniform statement about `specRatio` for the coupled kernel remains unproved. This is the substantive limitation, and it is a translation between two languages rather than a missing estimate — which does not make it small. Two further boundaries are worth naming. Theorem 5.2 discharges every hypothesis *except* Dobrushin’s condition, and whether that condition holds is a question about a particular interaction that this paper does not answer for any; the envelope of Theorem 3.1 is what a user would apply, and it is a sufficient condition, not a characterisation. And the decay proved is decay of correlations of a measure on a *finite* product space: no infinite-volume state is constructed anywhere, and the word *clustering* is accordingly not used for one.

10 Reproducibility

Anchor. All sources cited above are read at commit `4f3b864004c48edc87c09bd2537fe51340fb6411` of <https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME>. Toolchain `leanprover/lean4:v4.29.0-rc6`; Mathlib pinned at `07642720480157414db592fa85b626dafb71355b`. A runner verifies all three of these before compiling anything and aborts on any mismatch, so that a stale cache cannot masquerade as a reproduction.

Job counts, and how each delta was obtained. The development machine measured the core at 8465 jobs with Theorem 3.3’s module removed from the root and 8466 with it restored — that delta by rebuilding the same tree twice, not by subtracting against a baseline someone else measured. A fresh Linux clone reproduced 8466 at that commit. With the modules of Theorems 3.1 and 3.2 added, the Linux plane measures 8468 with zero errors: the +2 is the difference between two measurements on the same plane, one job per new module, and is not an extrapolation.

The four modules of Section 4 are *not* in the core, and the core count at the anchor is therefore still 8468. This is stated rather than left to be discovered: they are imported by neither `YangMillsCore` nor the repository oracle, so no repository-wide number in this paper covers them, and wiring them in — with the job-count delta measured against this campaign’s own parent commit, never against a merge-base — is separate work not done here.

Oracle. For Theorems 3.1–3.3, at an earlier anchor of this lane: the repository-wide oracle ran to completion with exit code zero over 2817 axiom reports and **zero** occurrences of `sorryAx`, thirty-seven of them this chain’s. For Theorems 4.3 and 5.2: two *targeted* oracle runs at this anchor, exit code zero, printing the axioms of seventeen and eleven endpoints respectively — among them the three clauses of Lemma 4.1, both Grüss forms with the Grüss attainment,

the transport domination, the finite- N and series estimates, the exponential-decay composition and the two-point corollary, and on the Gibbs side invariance, the zero diagonal, domination, the three instantiated endpoints and both witnesses. The union of every axiom list printed, over both runs, is exactly `{propext, Classical.choice, Quot.sound}`: Lean’s three standard axioms, no project axiom, no sorry in any build log. The union is computed as a *union* rather than by matching each printed line against the standard triple, because a declaration depending on *fewer* axioms is perfectly clean and would fail that naive test.

One gap in the runner’s own coverage, stated because a reader counting modules would otherwise not find it. The runner builds two lane modules by name; `DobrushinRowSum.lean` is not one of them. It is nevertheless imported by the core, so the zero exit of the core build covers it, and its declarations appear in the oracle.

Independent hash agreement. A second desk, working from its own check-out on a different operating system and without coordination on the value, reported `12c1fcdb12e39e46381fa372a102503f5d63522ecbae27c0b3911671a7c04eb7` at 11467 bytes for Theorem 3.3’s module. The Linux run computed the same digest and the same size. The oleans, the axiom reports and the elaboration therefore refer to the same bytes, which is the only reason any of the numbers above can be compared at all.

The bytes of Sections 4 and 5. Printed in the same executions that built and oracled them:

module	sha256	bytes
<code>DobrushinOscillation.lean</code>	<code>ca7c6842b8e08d13c405517994df86f868047b48c53fb9ab2e24fcea10777d89</code>	7777
<code>DobrushinGruss.lean</code>	<code>8e00f418a42fb30d6f6b367d5e353b4f34149c2e70d3625ae3307ce8bceb9ace</code>	16462
<code>DobrushinConditional.lean</code>	<code>eb52b816f2c401d17edb72c5a7d0f8d413c33bbf9292f770e92a787fc8f5833a</code>	18270
<code>DobrushinComparison.lean</code>	<code>a489fc2823e564685f3ffbf586f1a1f63ce4b78a955a9ec88edbcd70afae4133</code>	54738
<code>DobrushinGibbs.lean</code>	<code>b525731d274bf0ce7511e0e0f796d5892f22cee1601daa5a8fa7fea6bd507ba1</code>	16787

Certifiers. Four gate scripts, each run in `normal` and in `optimized` (`python -O`) mode on the Linux plane; all eight runs exited zero. `judge_dobrushin.py` reported 29 of 29 checks in each mode and `judge_dobrushin_d2.py` 18 of 18, at the previous anchor. At this anchor `judge_dobrushin_d3.py` reported 11 of 11 and `judge_dobrushin_d3b.py` 6 of 6, in both modes, and both ran *before* the build in the same runner invocation — the record shows the gates passing before the theorems they license were compiled. No acceptance decision in any of the four depends on a Python `assert`; each counts the checks it performed and refuses to print a verdict when fewer ran than expected, because an empty failure list is also what zero checks produce.

What the passes cost, and the one that was not a repair. Seven elaboration rounds on the Linux plane closed Section 4’s four modules. Not one of the failures was mathematical: they were library names absent under their classical spelling in the pinned Mathlib, section variables that a statement never mentioned and so never bound, a global rewrite that consumed the literal 1 inside an indicator, and goals equal only after β -reduction. The exception is worth its sentence. The first round failed in `DobrushinOscillation.lean`, a module committed at an *earlier* anchor: the file had been verified in its pre-split working form and split afterwards, and the committed bytes had never been elaborated. The lane had declared that debt in advance and the runner was built to re-verify those bytes; it found the defect on its first run. *A module that was green before an edit is not green after it, and the hash is what says which one was compiled.*

Child processes. Every child of the runner writes a sentinel holding exactly one line, its real decimal exit code, captured after termination and written atomically with validation. Four states are distinguished on read: absent, malformed, non-zero, zero. A run in which every sentinel reads zero is reported as a *pass candidate*, never as a pass, because the logs still have to be read.

Every module in the anchor compiles. An earlier version of this paper was anchored to a tree in which `DobrushinCoefficient.lean` did not, and said so in this section rather than leaving it to be discovered. It took five passes to close, and the record of those passes is worth one sentence because it is the honest description of what mechanising a classical argument costs: none of the fifteen errors across them was mathematical. The mechanism — the identity of Theorem 3.1 and the denominator bound under it — is the one written in the first draft and was never altered. Every failure was a library lemma whose name or form had been recalled wrongly, or a tactic emitting a different number of goals than the script assumed.

References

- [1] R. L. Dobrushin, *The description of a random field by means of conditional probabilities and conditions of its regularity*, Theory of Probability and its Applications **13** (1968), 197–224. The uniqueness condition and the comparison estimate whose mechanised form is Theorem 4.3. The theorems of this paper are classical and due here.
- [2] R. L. Dobrushin, *Prescribing a system of random variables by conditional distributions*, Theory of Probability and its Applications **15** (1970), 458–486. Together with [5], the source of the argument mechanised in Section 4: the single-site update acting on oscillation vectors, and the resolvent of the interdependence matrix as the resulting comparison.
- [3] G. Grüss, *Über das Maximum des absoluten Betrages von $\frac{1}{b-a} \int_a^b f(x)g(x) dx - \frac{1}{(b-a)^2} \int_a^b f(x) dx \int_a^b g(x) dx$* , Mathematische Zeitschrift **39** (1935), 215–226. The covariance bound with the quarter constant, applied pointwise to the conditional in Lemma 4.2. Our proof is the discrete signed-mass one rather than Grüss’s, so that each factor $\frac{1}{2}$ is created in exactly one lemma.
- [4] T. Popoviciu, *Sur les équations algébriques ayant toutes leurs racines réelles*, Mathematica (Cluj) **9** (1935), 129–145. The variance bound $(M - m)^2/4$, machine-checked in the same module as Lemma 4.2 and deliberately *not* used by it.
- [5] B. Simon, *The Statistical Mechanics of Lattice Gases, Volume I*, Princeton University Press, 1993. Chapter V gives the Dobrushin uniqueness condition, the interdependence matrix, and the exponential decay of correlations it yields; the one-bond envelope $\tanh |J|$ for the Ising cell and the resulting window $2 \tanh \beta + 2 \tanh \gamma < 1$ are the standard specialisation.
- [6] L. Onsager, *Crystal statistics I: a two-dimensional model with an order-disorder transition*, Physical Review **65** (1944), 117–149. The exact solution, and the source of the line $\sinh 2\beta \sinh 2\gamma = 1$ used as the reference boundary in Section 6.
- [7] B. Kaufman, *Crystal statistics II: partition function evaluated by spinor analysis*, Physical Review **76** (1949), 1232–1243. The transfer-matrix spectrum in closed form. This is the reason Section 8 claims generality rather than constants: for this cell the exact second eigenvalue is available and the Dobrushin constant is worse than it.

- [8] I. Schur, *Bemerkungen zur Theorie der beschränkten Bilinearformen mit unendlich vielen Veränderlichen*, Journal für die reine und angewandte Mathematik **140** (1911), 1–28. The test whose constant-row-sum form is the method discussed in Section 2.
- [9] E. Seneta, *Non-negative Matrices and Markov Chains*, revised printing, Springer, 2006. Standard reference for nonnegative matrices, ergodicity coefficients and the resolvent series $\sum_n C^n = (I - C)^{-1}$ of which Theorem 3.3 is a locally-supported, entrywise refinement.
- [10] The mathlib Community, *The Lean mathematical library*, Proceedings of CPP 2020, 367–381. The library against which the development is pinned; it carries a formalisation of the Perron–Frobenius theorem for irreducible nonnegative matrices.
- [11] *A Machine-Verified Proof of a Quantum-Optimization Conjecture*, arXiv:2606.29687 (2026). A Lean 4 formalisation of the Ising chain with momentum-mode decomposition and the Jordan–Wigner transformation. It concerns the transverse-field quantum chain rather than the classical two-dimensional transfer matrix, and displaces nothing here; it is cited because a blanket claim that the Ising model has not been formalised would be false.
- [12] *Formalization of Quantum Field Theory*, arXiv:2603.15770 (2026). Osterwalder–Schrader axioms for the Gaussian free field in Lean 4; the nearest mechanised neighbour on the constructive side, and free rather than interacting.
- [13] *Spectrum and reflection positivity on a finite lattice*, arXiv:2606.07922 (2026). Finite-lattice spectral and reflection-positivity results for a related model, not mechanised.